

ES -116

Principles and Applications of Electrical Engineering

Semester II, 2025-26

Resistive Network Analysis

Recap: Fundamental Laws in Electrical Circuits

- Conservation of Charge → **The net current entering a node is zero (Kirchhoff's current law).**
- Conservation of Energy → **The algebraic sum of the voltages for any loop equals zero (Kirchhoff's voltage law).**

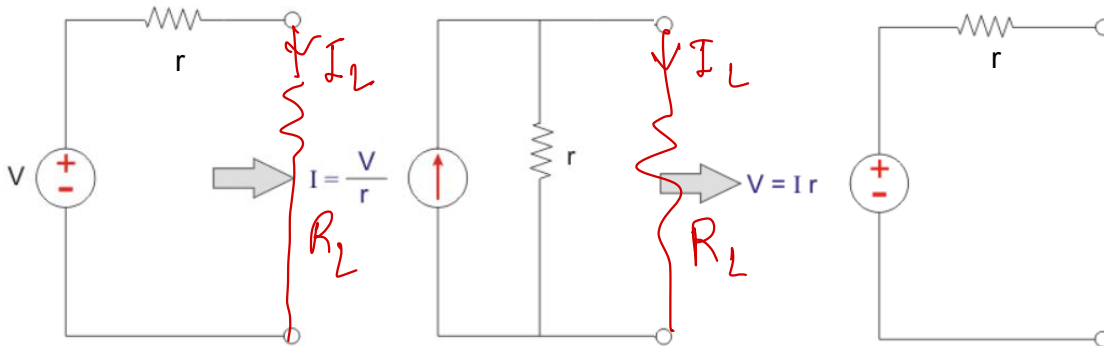
$$\sum_{n=1}^N i_n = 0 \quad \text{Kirchhoff's current law}$$

- Gives us a relation for equivalent resistance between two nodes for parallel resistive branches.
- Enables us to sum the currents from current sources in different branches between two nodes.

$$\sum_{n=1}^N v_n = 0 \quad \text{Kirchhoff's voltage law}$$

- Gives us a relation for equivalent resistance between two nodes for series resistive branches.
- Enables us to sum the voltages from voltage sources in a branch between two nodes.

Voltage/Current Source Transformation



Can we prove that it applies even if there is a load connected between the two nodes?

→ We can calculate the current through the load for both equivalent circuits.

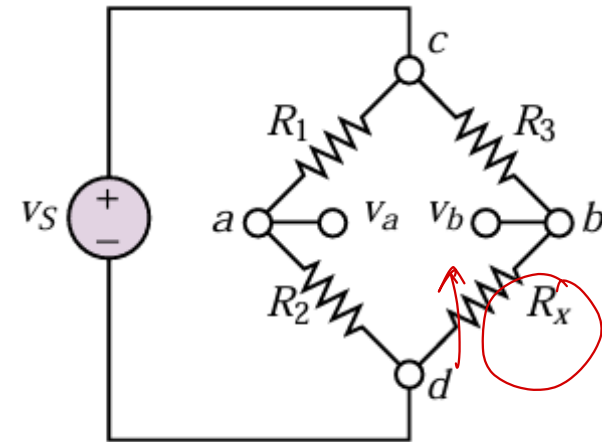
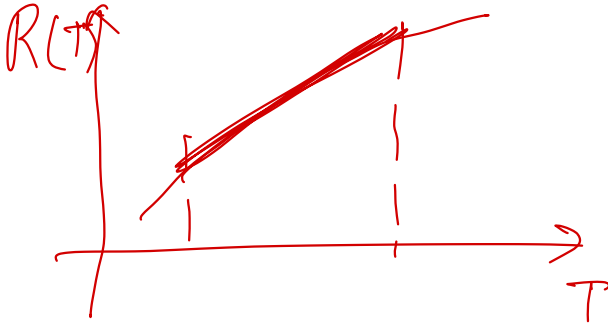
Question

Q. A medical lab wants an electronic thermometer around 37°C using a thermistor (whose resistance changes with temperature).

They need a DC circuit that:

- Gives zero output at 37°C .
- Produces a small voltage that changes clearly when temperature moves above or below 37°C .

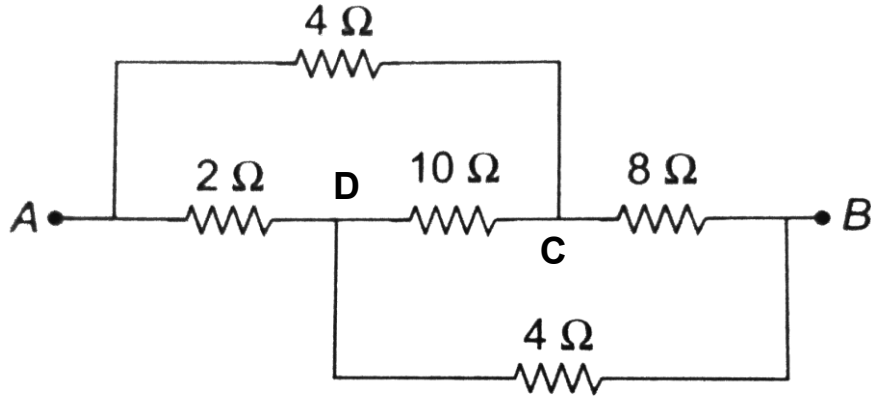
Design a resistive network using the thermistor and fixed resistors to meet this (i.e. calculate R_x so that v_{ab} is zero).



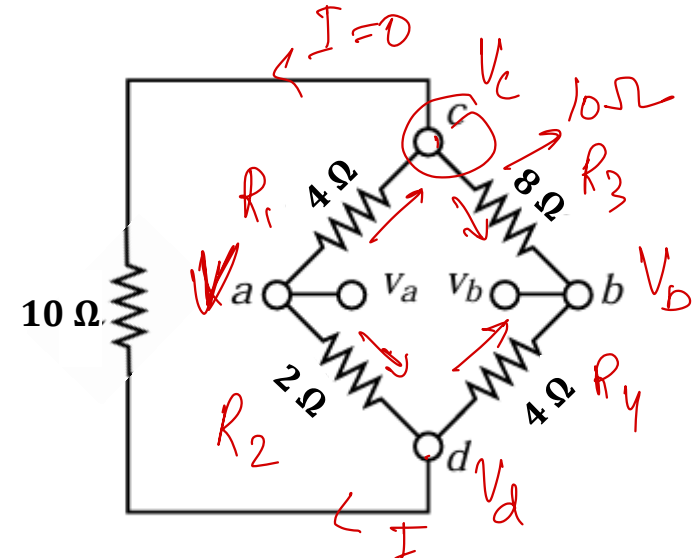
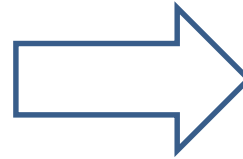
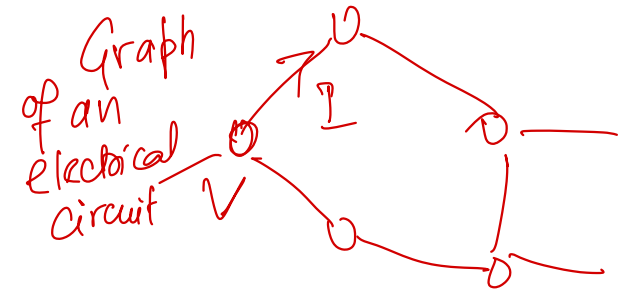
- Similar to a Question in Tutorial 1.

Problem from Lecture 3

Calculate R_{AB} .



nodes & branches



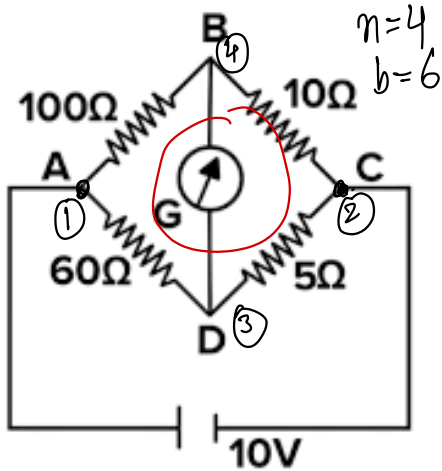
What if it was an unbalanced bridge?

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$\text{or } \frac{R_1}{R_3} = \frac{R_2}{R_4}$$

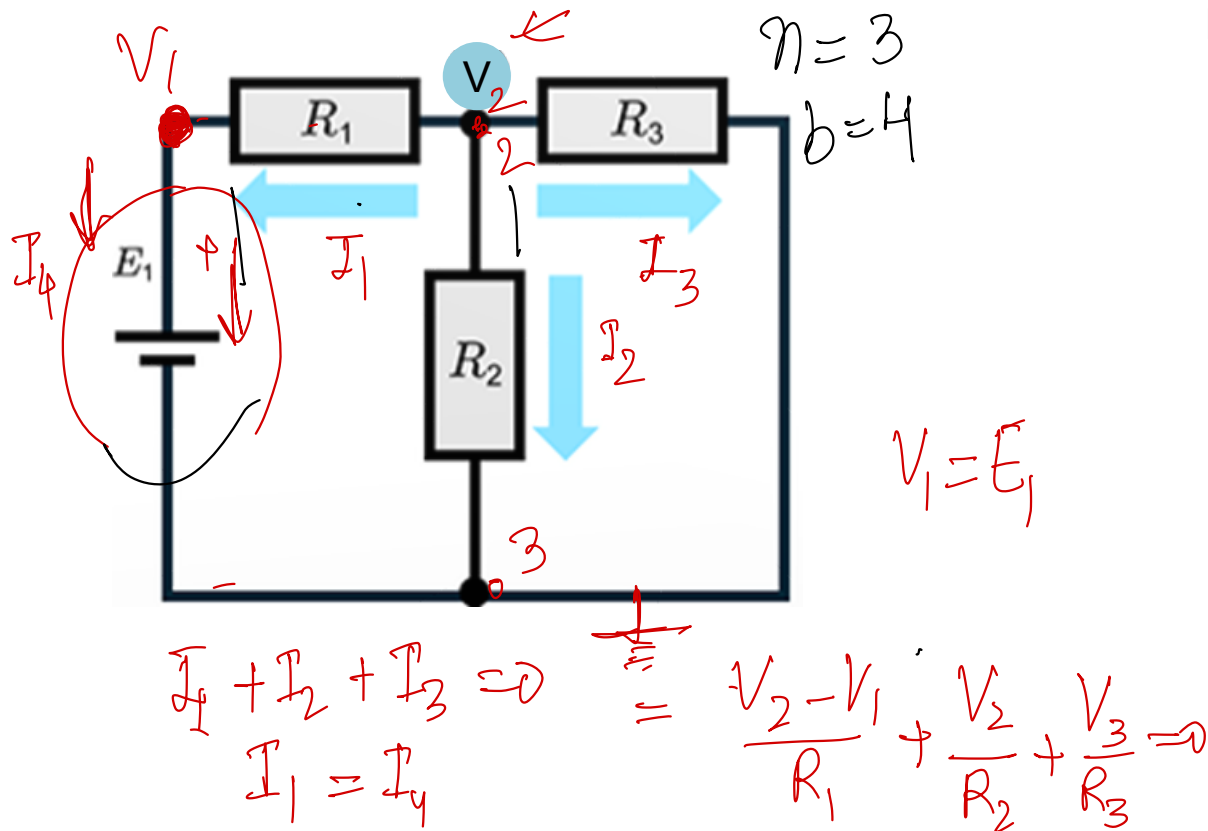
Why do we need structured way to analyse circuits?

How would you calculate the current flowing through Galvanometer?

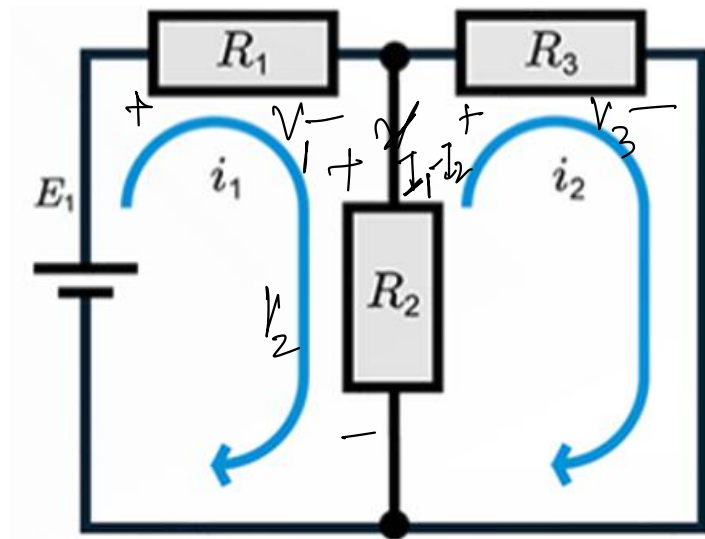


Nodal Analysis → Voltages
Mesh Analysis → Currents flowing in meshes

Nodal and Mesh/Loop Analysis Methods



Mesh: Special loop that doesn't contain any loops within it.

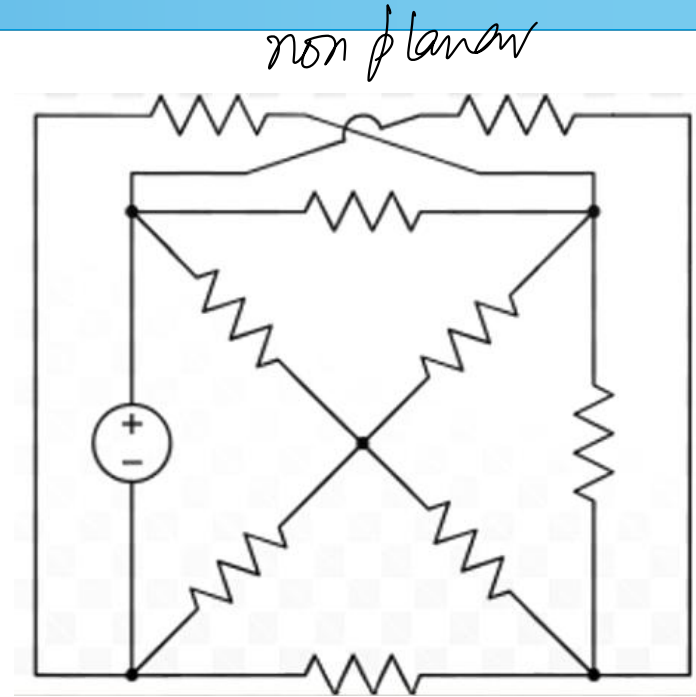
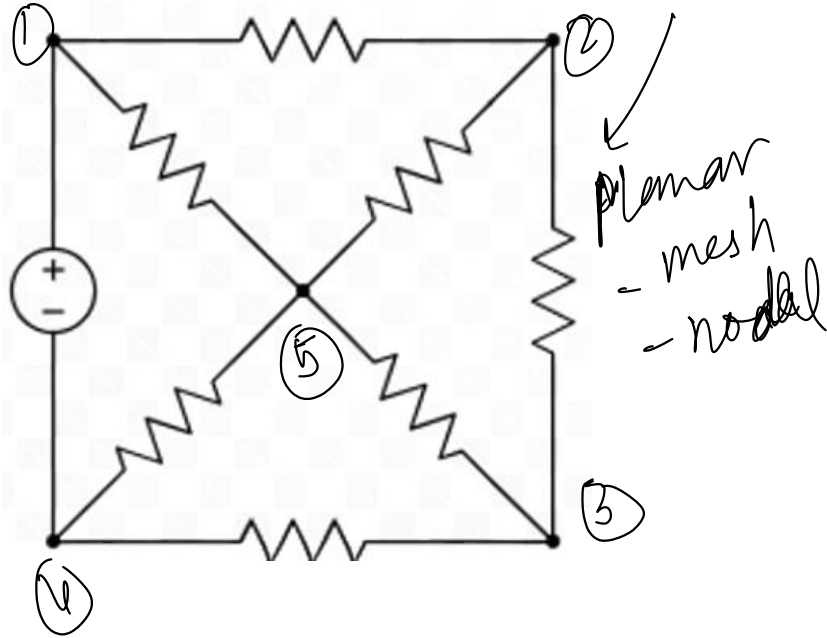


$$\begin{aligned}
 E_1 - V_1 - V_2 &= 0 \Rightarrow E_1 - I_1 R_1 - (I_1 - I_2) R_2 = 0 \\
 -V_3 + V_2 &= 0 \\
 -I_2 R_3 + (I_1 - I_2) R_2 &= 0
 \end{aligned}$$

- Branch is the path between two nodes that contains exactly one two-terminal element.
- Node is the connection point where the terminals of two or more elements meet and share the same potential.

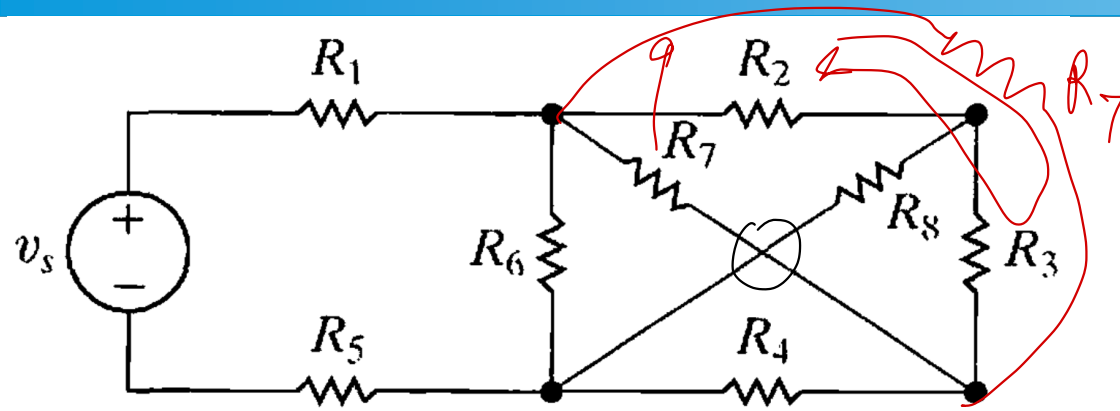
Note about planar vs non-planar circuits

- Can you apply nodal/mesh analysis in both circuits?



nodal
Analysis

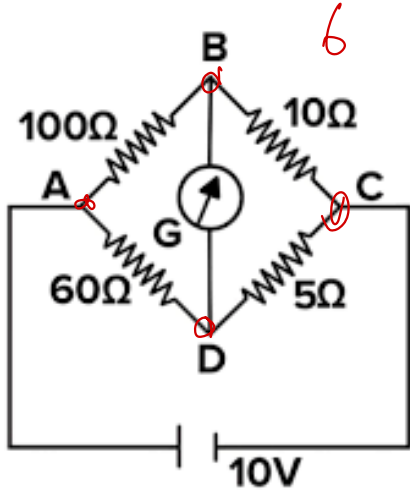
Q. Can Mesh Analysis be applied to the given circuit?



- Check if it can be drawn on a plane with no crossing branches!

Nodal/Mesh Analysis: Number of Nodes vs Meshes

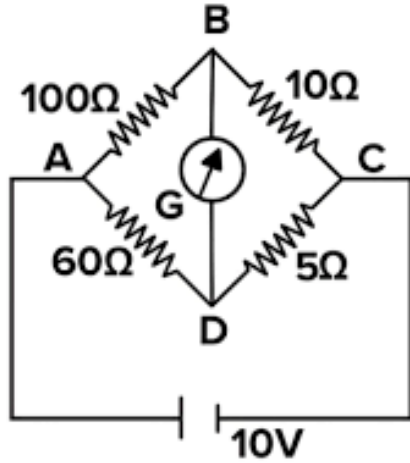
How would you calculate the current flowing through Galvanometer?



$$\begin{aligned} n, b \\ m &= b - n + 1 \\ &= b - (n - 1) \end{aligned}$$

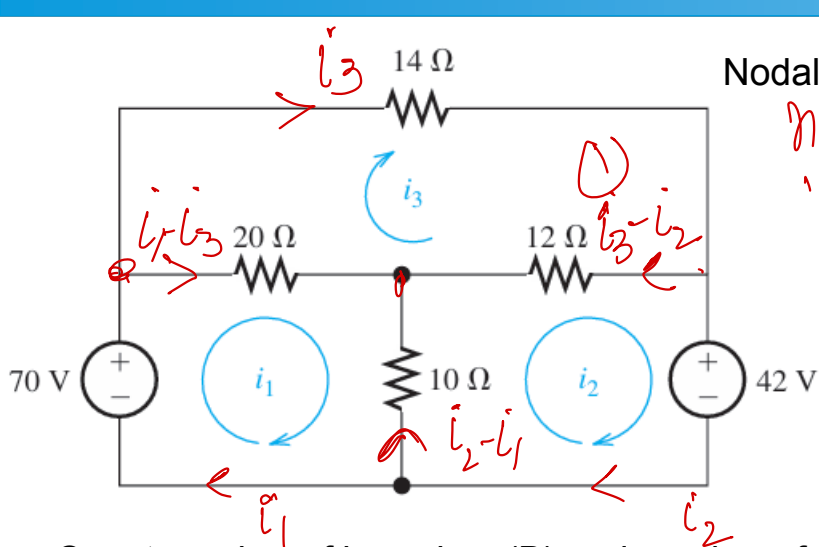
How many Simultaneous Equations?

To generalize, let us represent "n": number of nodes, b: number of branches, how many mesh will you have?



$$\text{Number of mesh} = b - n + 1$$

Q. Solve for the current in each element of the circuit



Count number of branches (B) and number of nodes (N):
If $B - N + 1 > N - 1$, go for nodal, otherwise mesh

Nodal/Mesh?

$$n = 4$$

$$= 6$$

meshes
 $b - n + 1 = 3$
~~nodes~~
 $n - 1 = 3$

$i_1 \quad i_2 \quad i_3$
mesh analysis

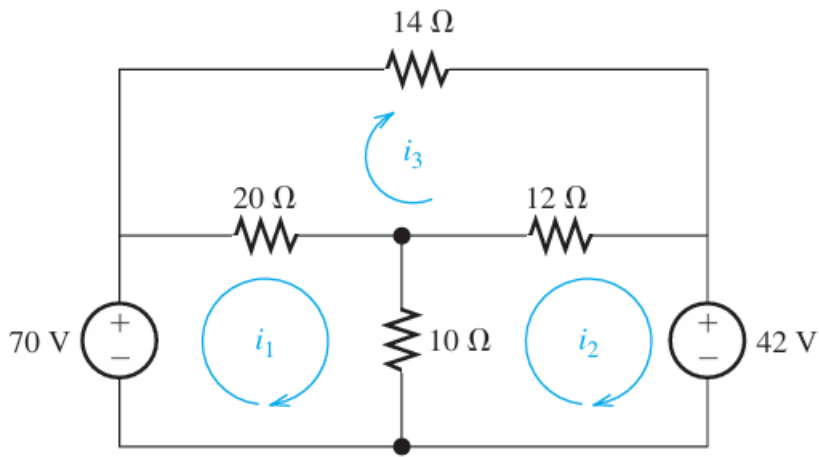
mesh 1: $70 - (i_1 - i_3)20 + (i_2 - i_1)10 = 0$
 $\Rightarrow -30i_1 + 10i_2 + 20i_3 = -70$

mesh 3: $-14i_3 - 12(i_3 - i_2) + (i_1 - i_3)20 = 0$
 $\Rightarrow 20i_1 + 12i_2 - 46i_3 = 0$

mesh 2: $12(i_3 - i_2) - 42 - (i_2 - i_1)10 = 0$
 $\Rightarrow 10i_1 - 22i_2 + 12i_3 = 42$

$\rightarrow$ 3 variables & 3 linear equations to solve
 for i_1, i_2 & i_3 $\rightarrow$ node voltages can be
 determined using ohm's Law.

Writing it in the Matrix Format



$G = R^{-1}$
conductance matrix

$$[I] = [G][V]$$

3×1 3×3 3×1

$$\begin{bmatrix} -30 & 10 & 20 \\ 10 & -22 & 12 \\ 20 & 12 & -46 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -70 \\ 42 \\ 0 \end{bmatrix}$$

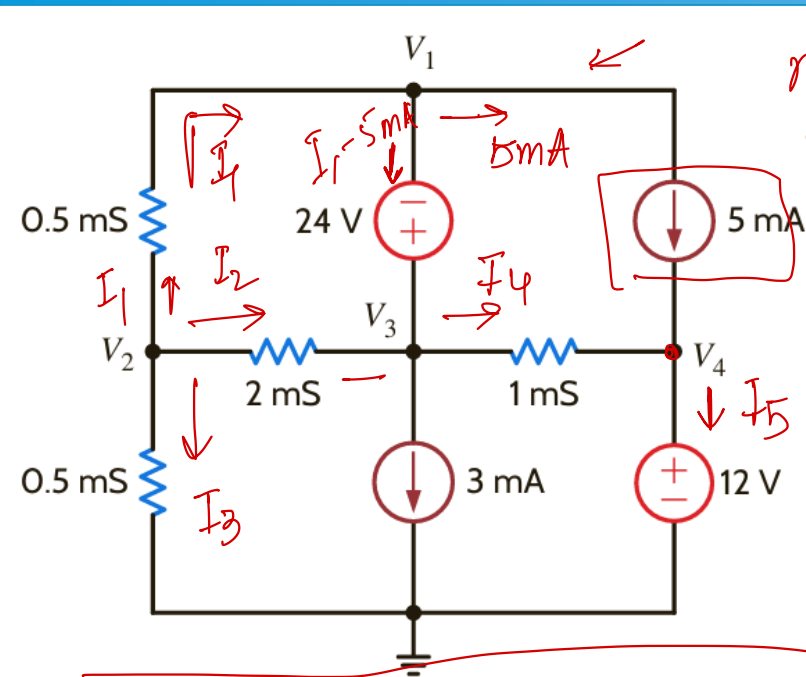
$$\begin{bmatrix} R \end{bmatrix} \begin{bmatrix} I \end{bmatrix} = \begin{bmatrix} V \end{bmatrix}$$

3×3 3×1 3×1

You may use MATLAB to solve it:

```
>> R = [      ;      ;      ]
>> V = [      ;      ;      ]
>> I = inv(R)*V
```

Another Example: Calculate node voltages



$n=5$
 $n-1$ equations
 $V_4 = 12V$, at node 3

First calculating using Nodal Analysis

$$\begin{aligned} I_2 + I_1 - 5 \text{ mA} &= 3 \text{ mA} + I_4 \\ 2 \text{ m} (V_2 - V_3) + (V_2 - V_1) 0.5 \text{ m} &= 8 \text{ mA} + (V_3 - V_4) 1 \text{ m} \\ \Rightarrow 2V_2 - 2V_3 + 0.5V_2 - 0.5V_1 &= 8 + V_3 - 12 \\ \Rightarrow -0.5V_1 + 2.5V_2 - 3V_3 &= -4 \quad \text{--- (1)} \end{aligned}$$

at Node 2 $I_1 + I_2 + I_3 = 0$

$$\begin{aligned} (V_2 - V_1) 0.5 \text{ m} + (V_2 - V_3) 2 \text{ m} + V_2 \cdot 0.5 \text{ m} &= 0 \\ \Rightarrow 0.5V_2 - 0.5V_1 + 2V_2 - 2V_3 + 0.5V_2 &= 0 \\ \Rightarrow -0.5V_1 + 3V_2 - 2V_3 &= 0 \quad \text{--- (2)} \end{aligned}$$

Node 4

$$\begin{aligned} I_5 &= I_4 + 5 \text{ mA} \\ -I_3 - 3 \text{ mA} &= I_4 + 5 \text{ mA} \\ I_3 + I_4 + 8 \text{ mA} &= 0 \end{aligned}$$

$$\begin{aligned} V_2 \cdot 0.5 \text{ m} + (V_3 - 12) \cdot 1 \text{ m} + 8 \text{ mA} &= 0 \\ \Rightarrow 0.5V_2 + V_3 - 12 + 8 &= 0 \Rightarrow 0.5V_2 + V_3 = 4 \quad \text{--- (3)} \end{aligned}$$

nodes $V_3 - V_1 = 24 \quad \text{--- (4)}$

$$V_3 = V_1 + 24$$

$$(1) \rightarrow -0.5V_1 + 2.5V_2 - 3V_1 - 72 + 4 = 0$$

$$\Rightarrow -3.5V_1 + 2.5V_2 = 68$$

$$(3) \rightarrow 0.5V_2 + V_1 + 24 = 4$$

$$\Rightarrow V_1 + 0.5V_2 = -20$$

$$5V_1 + 2.5V_2 = -100$$

$$-3.5V_1 + 2.5V_2 = 68$$

+

-

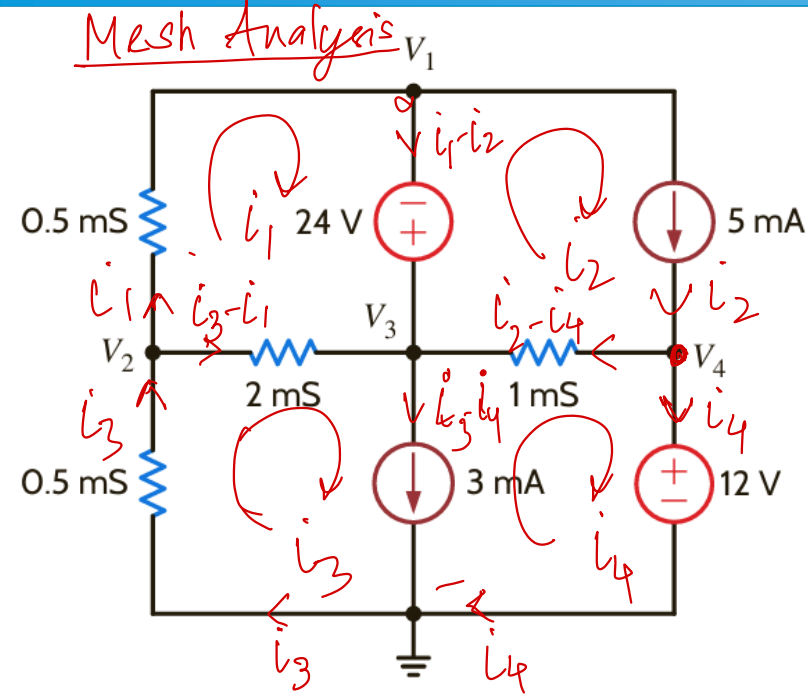
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$$8.5V_1 = -168$$

$$\Rightarrow V_1 = -19.76$$

Another Example: Calculate node voltages

Mesh Analysis



$R = \frac{1}{G}$ i.e. $0.5 \text{ mS} \rightarrow 2 \text{ K}$
 $2 \text{ mS} \rightarrow 5 \text{ K}$
 $1 \text{ mS} \rightarrow 1 \text{ K}$

$$i_2 = 5 \text{ mA}, \quad i_3 - i_4 = 3 \text{ mA}, \quad V_4 = 12 \text{ V}$$

$$(i_2 - i_4) 1 \text{ K} - 12 + V_3 = 0 \quad \text{--- (1)}$$

$$-i_3 \cdot 2 \text{ K} - (i_3 - i_1) \cdot 0.5 \text{ K} - V_3 = 0 \quad \text{--- (2)}$$

$$-i_1 \cdot 2 \text{ K} + 24 + (i_3 - i_1) \cdot 0.5 \text{ K} = 0 \quad \text{--- (3)}$$

(1) + (2) $\Rightarrow$ if i are in mA,

$$i_2 - i_4 - 12 - 2i_3 - 0.5i_3 + 0.5i_1 = 0$$

$$\Rightarrow 0.5i_1 + i_2 - 2.5i_3 - i_4 = 12 \quad \text{--- (A)}$$

$$i_2 = 5 \quad \text{--- (B)}$$

$$i_3 - i_4 = 3 \quad \text{--- (C)}$$

$$(3) \Rightarrow -2.5i_1 + 0.5i_3 = -24 \quad \text{--- (D)}$$

$$\textcircled{A} - \textcircled{C} - \textcircled{B}$$

$$\Rightarrow 0.5 \dot{I}_1 - 3.5 \dot{I}_3 = 12 - 5 - 3$$

$$= 4$$

$$\Rightarrow 2.5 \dot{I}_1 - 17.5 \dot{I}_3 = 20$$

$$- 2.5 \dot{I}_1 + 0.5 \dot{I}_3 = -24$$

$$\underline{\hspace{10em}} -17 \dot{I}_3 = -4 \Rightarrow \dot{I}_3 = \frac{4}{17} \text{ mA}$$

$$\dot{I}_1 = \left(4 + \frac{3.5 \times 4}{5 + 17} \right) 2 = \frac{48}{5} \text{ mA}$$

②

$$\rightarrow V_3 = 0.5 \dot{I}_1 - 2.5 \dot{I}_3 = \frac{24}{5} - \frac{2.5 \times 4}{17} = \frac{24}{5} - \frac{10}{17} = 4.2117 \text{ V}$$

$$V_1 = V_3 - 24 = -19.788 \text{ V}$$

Summary

- Both methods are systematic ways to apply KCL/KVL to complex electrical networks.
- Nodal analysis
 - Unknowns: node voltages ($\approx n-1$ equations).
 - Based on KCL at each essential node.
 - Especially convenient with current sources and for non-planar circuits.
- Mesh analysis
 - Unknowns: mesh currents ($\approx b-n+1$ equations, planar only).
 - Based on KVL around each mesh.
 - Especially convenient when the circuit is planar and has mainly voltage sources.

Choosing a Method

- Count equations:
 - Nodal unknowns $\approx n-1$; mesh unknowns $\approx b-n+1$.
 - Pick the method with fewer equations that directly gives what you want (voltages $\rightarrow$ nodal, currents $\rightarrow$ mesh).
- Remember:
 - Nodal works for any network; mesh requires a planar circuit.
 - Both ultimately give the same currents and voltages; they are just different viewpoints on the same physics.

If time permits

What is peculiar about these equations?

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix}$$

$$[R][I] = [V] \text{ or } [I] = [G][V]$$

Set of linear equations

Why? → Because we are analyzing linear circuits.

Linear circuits: Circuits consisting of linear components (e.g. resistors).

Properties of linear circuits:

1) Homogeneity (Scaling): If you scale the input sources by factor k , the output response is also scaled by the same factor k .

E.g. $I = GV$, If you triple V , I triples. While with a non-linear device, where $I = I_0 \exp(\frac{V}{V_0})$, it does not apply.

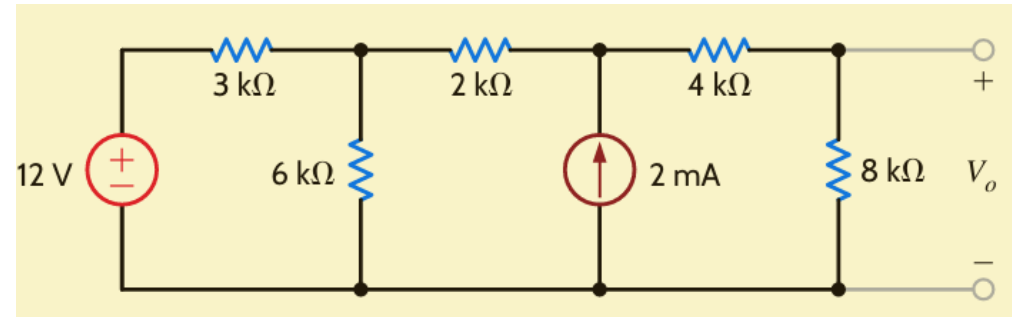
2) Additivity (Superposition): The response to a sum of inputs is the sum of the individual responses.

E.g.

Linear circuit: $f(x_1+x_2) = f(x_1) + f(x_2)$

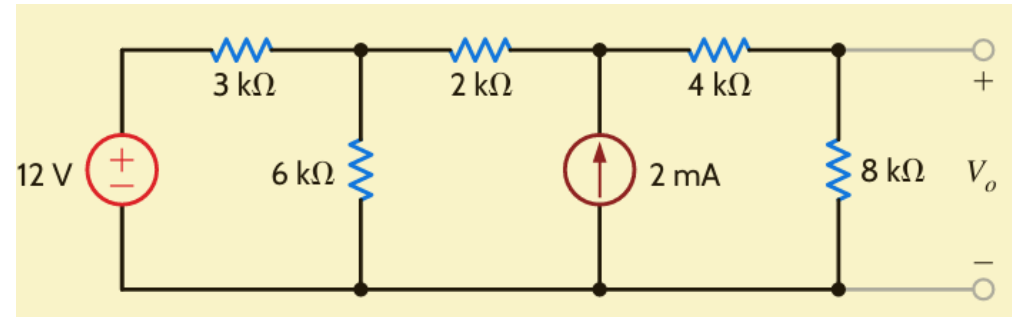
Non-linear circuit: $I = kV^2$, If $V_1 = 1 \text{ V}$, $V_2 = 1 \text{ V}$, it doesn't add up.

Let's put linearity to the test - I



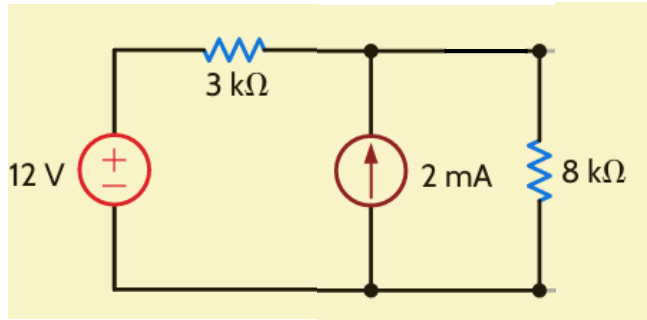
- We solved this problem using repeated source transformation.
- Let us try to work it backwards, i.e. assuming V_o value and finding the source value and then use the homogeneity property (scaling) of linear circuits.

Let's put linearity to the test - II



- We solved this problem using repeated source transformation.
- Let us try to use Superposition property of linear circuits to solve this problem

Superposition Theorem/Method

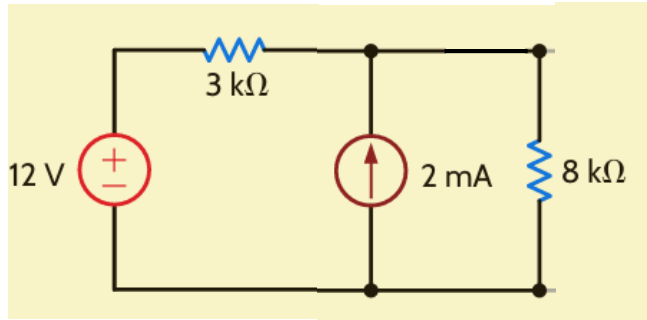


Calculate the current through 8 Ω load resistance.

Try through breaking it in two networks

- The current through, or voltage across, any device is equal to the algebraic sum of the currents or voltages produced independently by each source.
- It allows us to find a solution for a current or voltage using only one source at a time.
- To consider the effects of each source, the other sources must be removed.
- When removing a voltage source from a network schematic, replace it with a short circuit i.e. zero ohms. Not the internal resistance.
- When removing a current source from a network schematic, replace it by an open circuit of infinite ohms. Not the internal resistance.
- Since the effect of each source will be determined independently, the number of networks to be analysed will equal the number of sources.

Superposition Theorem/Method



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